

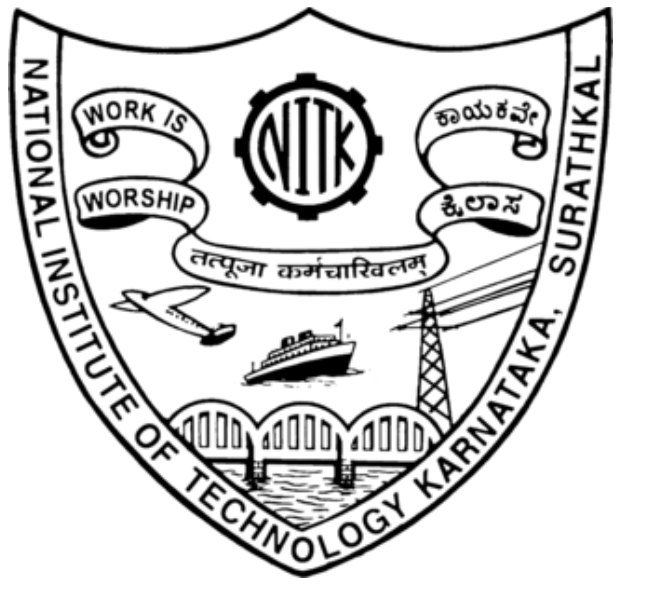


Integration of BIM and GIS for a Campus Building Model

Professional Association For Civil Engineering, NITK Surathkal

Mentor: Amarnath , Asiya

Mentees: Vismaya, Kalyani, Hans, Navdeep, Joseph



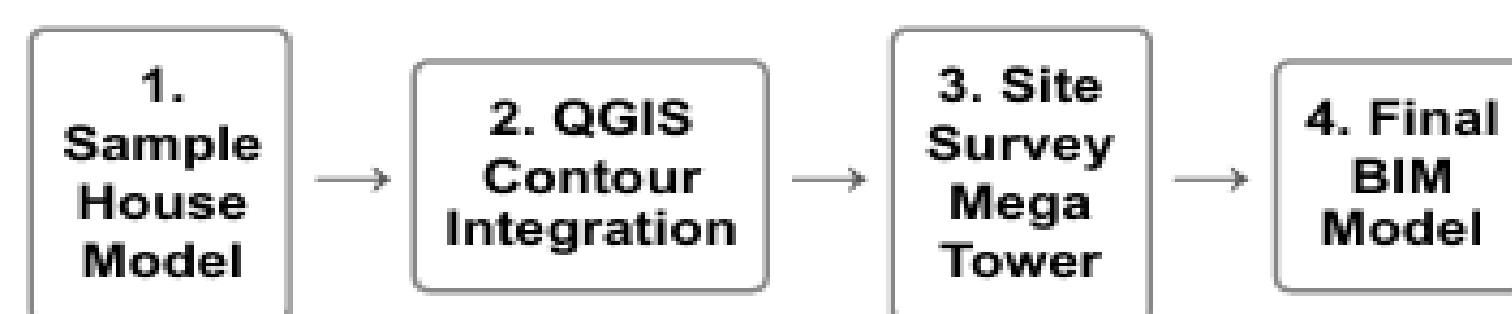
Introduction

- Context: The rapid advancement of digital technologies in the AEC (Architecture, Engineering & Construction) industry has made BIM and GIS powerful tools for building documentation and spatial analysis. Integrating these platforms enables more accurate, data rich models of real-world structures.
- Project Objective: To create a detailed 3D Building Information Model of the Mega Tower (Himalaya Block), NITK Surathkal, and integrate it with GIS-based terrain data for comprehensive spatial visualization.
- Methodology: The project utilizes Autodesk Revit for BIM modeling, QGIS for geospatial data processing, and Twinmotion for real-time rendering and visualization.

Scope

- Learning and applying Autodesk Revit for architectural 3D modeling through a sample house prototype.
- Processing and integrating campus terrain/contour data using QGIS for accurate site placement.
- Conducting a detailed physical survey of the Mega Tower for precise dimensional accuracy.
- Developing a complete BIM model of the Himalaya Block with GIS terrain integration. Producing photorealistic visualizations of the final model using Twinmotion.

Workflow



Software Used

Revit: Primary BIM authoring tool for 3D architectural modeling of the Mega Tower.

QGIS: Open-source GIS platform for processing and visualizing campus contour and terrain data.

Twinmotion: Real-time rendering engine for photorealistic visualization of the final model.

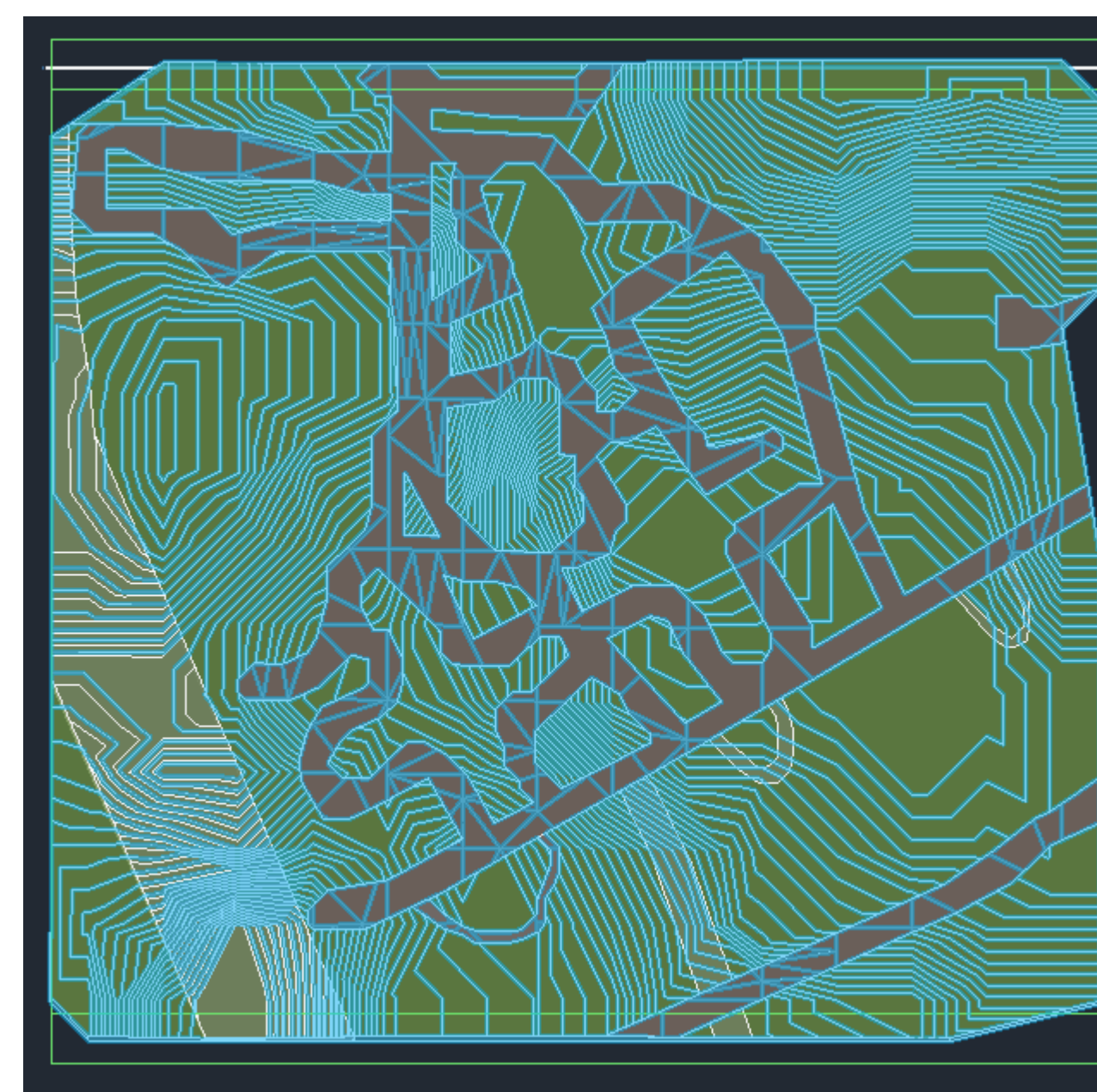
Numerical Analysis

Building Specifications

Building Name	Mega Tower (Himalaya Block)
Location	NITK Surathkal, Karnataka
Building Type	Institutional / Academic
Primary BIM Tool	Autodesk Revit
GIS Platform	QGIS
Visualization Tool	Twinmotion
Terrain Data Source	Campus Contour Map
Survey Method	Manual Field Measurement

Phase 1: Learning Revit — Sample House Model The project commenced with an introductory exercise in Autodesk Revit by modeling a sample residential structure. This phase focused on mastering fundamental BIM concepts — wall systems, floor plates, roof geometry, door/window families, and basic MEP elements — establishing the technical foundation for the more complex Mega Tower model.

Phase 2: GIS Integration — QGIS Contour Mapping Campus terrain data was processed in QGIS to generate accurate contour maps of NITK Surathkal. The geospatial data was then linked with the Revit environment to provide real world topographic context, enabling accurate site placement and terrain aware design decisions for the building model.



Phase 3 — Final BIM Mega Tower Model, survey accurate 3D model of Himalaya Block with GIS terrain and Twinmotion renders

Results and Discussion

GIS-BIM Integration — Contour Analysis: The QGIS-derived contour map of the NITK campus was successfully processed and integrated with the Revit environment. This integration provides accurate topographical context for the building model, enabling site-aware design and spatial analysis.

Field Survey — Dimensional Accuracy: The physical survey of the Mega Tower yielded precise room-by-room measurements across all floors. Key dimensions documented include corridor widths, structural bay spacings, staircase geometry, and room configurations — all critical for ensuring the BIM model faithfully represents the actual building.

BIM Model — Mega Tower (Himalaya Block): A detailed 3D architectural model of the Himalaya Block was developed in Autodesk Revit using field survey data. The model accurately captures the floor layout, structural grid, room configurations, and building envelope, serving as a reliable digital twin of the physical structure.

Photorealistic Visualization — Twinmotion: Twinmotion was employed to render the final Revit model with photorealistic materials, lighting, and environmental context. The renders provide a compelling visual representation of the Mega Tower within its campus setting, useful for documentation, presentations, and future infrastructure planning

Conclusions

The Mega Tower (Himalaya Block) at NITK Surathkal was successfully modeled using Autodesk Revit, with campus terrain data integrated through QGIS to provide accurate spatial and topographic context. The project demonstrates the practical effectiveness of combining BIM and GIS workflows for institutional campus infrastructure documentation.

The structured four-phase workflow — commencing with foundational Revit learning, progressing through GIS terrain integration and detailed physical survey, and culminating in the final BIM model and Twinmotion visualization — proved essential in building technical proficiency and delivering a high-fidelity digital representation of the structure.

The photorealistic Twinmotion renders further enhance the model's utility for visualization, stakeholder communication, and future infrastructure development planning at NITK Surathkal. This project establishes a replicable methodology for BIM-GIS integration applicable to other campus buildings.